

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

Square P has a side length of x inches. Square Q has a perimeter that is **176** inches greater than the perimeter of square P. The function f gives the area of square Q, in square inches. Which of the following defines f ?

Answer

- A. $f(x) = (x + 44)^2$
- B. $f(x) = (x + 176)^2$
- C. $f(x) = (176x + 44)^2$
- D. $f(x) = (176x + 176)^2$

Correct Answer: A

Rationale

Choice A is correct. Let x represent the side length, in inches, of square P. It follows that the perimeter of square P is **4x** inches. It's given that square Q has a perimeter that is **176** inches greater than the perimeter of square P. Thus, the perimeter of square Q is **176** inches greater than **4x** inches, or **4x + 176** inches. Since the perimeter of a square is **4** times the side length of the square, each side length of Q is $\frac{4x+176}{4}$, or $x + 44$ inches. Since the area of a square is calculated by multiplying the length of two sides, the area of square Q is $(x + 44)(x + 44)$, or $(x + 44)^2$ square inches. It follows that function f is defined by $f(x) = (x + 44)^2$.

Choice B is incorrect. This function represents a square with side lengths $(x + 176)$ inches.

Choice C is incorrect. This function represents a square with side lengths $(176x + 44)$ inches.

Choice D is incorrect. This function represents a square with side lengths $(176x + 176)$ inches.